

ARTIFICIAL INTELLIGENCE

Assignment - 1

AI-Based Smart Classroom Attendance System

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AI-Based Smart Classroom Attendance System

1 Introduction

The AI Smart Attendance Agent v2.0 is a Python-based intelligent software agent developed to simulate an AI-powered classroom attendance system. The system initializes several AI modules, including the camera module, facial recognition engine, attendance database, and decision-making module. The system also prevents duplicate attendance entries and rejects unknown students, making the attendance process more organized and reliable.

2 PEAS Representation

Performance Measure	Environment	Actuators	Sensors
High recognition accuracy, prevention of duplicate attendance, fast processing, reliable attendance records, secure student identification	Classroom, students, attendance database, teacher, computer system	Display recognition result, record attendance in CSV, generate attendance dashboard, reject unauthorized users	Simulated facial recognition input, student database (JSON), AI confidence evaluation, liveness verification, mask detection

Table 1: PEAS Representation of the AI-Based Smart Classroom Attendance System

3 Environment properties

Property	Classification	Justification
Observability	Partially Observable	The AI agent relies only on the detected face and student database.
Determinism	Stochastic	Confidence score, mask detection, and liveness are generated through AI simulation.
Episodic / Sequential	Sequential	Every attendance decision updates the attendance database and affects future duplicate checks.
Static / Dynamic	Dynamic	Students enter the classroom continuously while attendance is being recorded.
Discrete /Continuous	Continuous	Each student recognition is processed as a separate event.
Single / Multi-Agent	Single-Agent	Only one AI attendance agent performs all recognition and attendance operations.

Table 2:Environment Properties of the AI-Based Smart Classroom Attendance System

4 Rationality Measure

The AI Attendance Agent acts rationally by making intelligent decisions before marking attendance.

Initially, the system scans the classroom and detects a student's face. It searches the student database to determine whether the detected student is registered. When a match is found, the AI module generates a confidence score between 95% and 99.9%, performs a simulated liveness verification, and checks for mask detection.

Attendance is marked only if the student exists in the database and has not already been marked during the current session. Unknown students are rejected immediately. Duplicate attendance entries are prevented by maintaining a list of students whose attendance has already been recorded. These decision-making steps ensure that attendance records remain accurate, secure, and reliable.

5 Innovation Point

The system generates an AI confidence score for every recognized student during attendance verification. This score indicates how accurately the student's identity matches the stored database. Attendance is accepted only

when the confidence level is sufficiently high. This improves the reliability of the attendance process and minimizes recognition errors.

6.Diagram

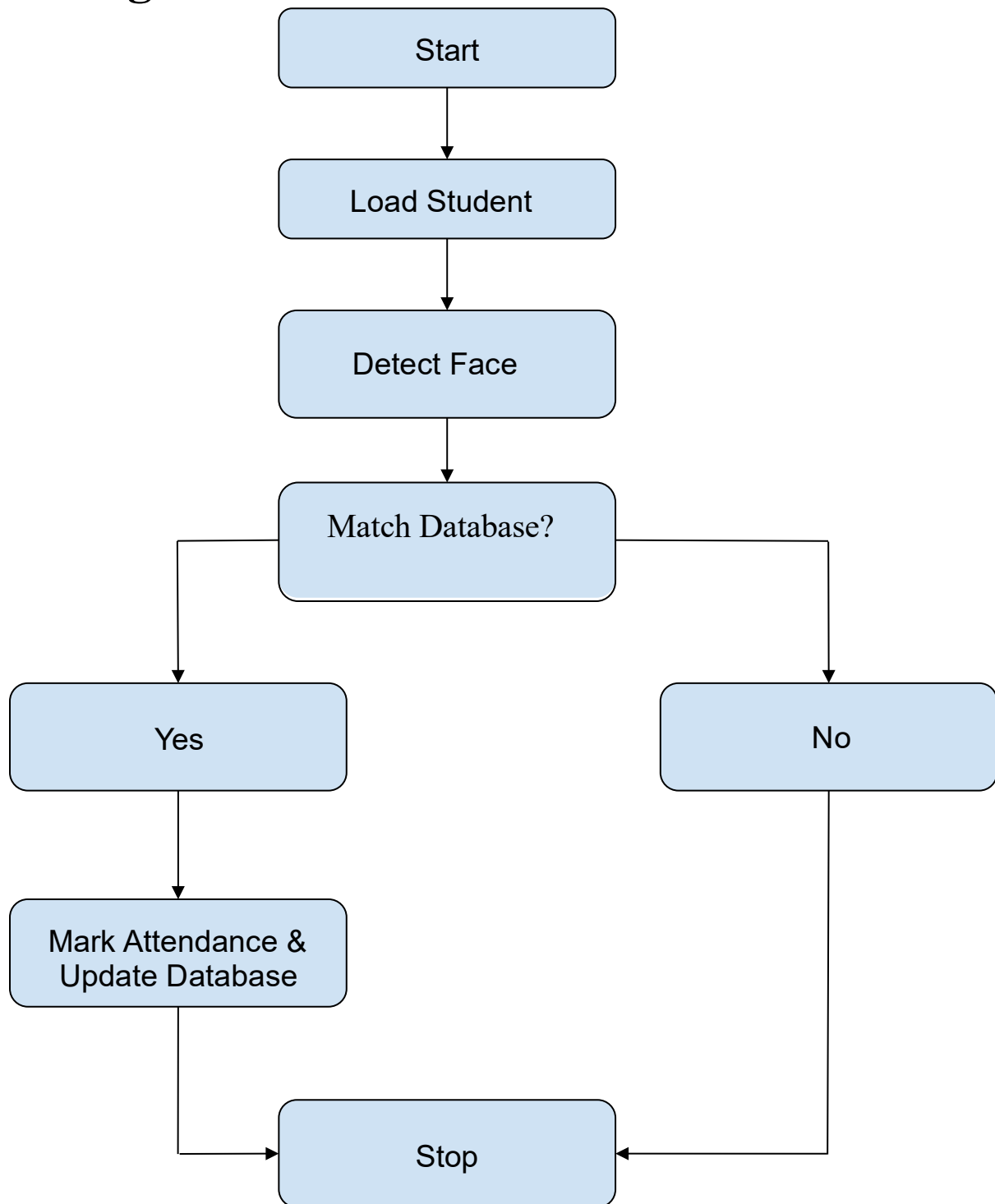


Figure 1: Architecture of the AI-Based Smart Examination Hall Seating System

7. Conclusion

The AI Smart Attendance Agent v2.0 demonstrates how an intelligent software agent can automate classroom attendance management. The system integrates student identification, confidence evaluation, liveness verification, duplicate attendance prevention, and attendance reporting into a single workflow.

Although the current implementation is a simulation, it provides a strong foundation for future integration with real-time facial recognition technologies such as OpenCV and machine learning models. The proposed system reduces manual effort, improves attendance accuracy, and enhances classroom management through intelligent decision-making.

8. Simulation

The project was implemented as a Python console application.

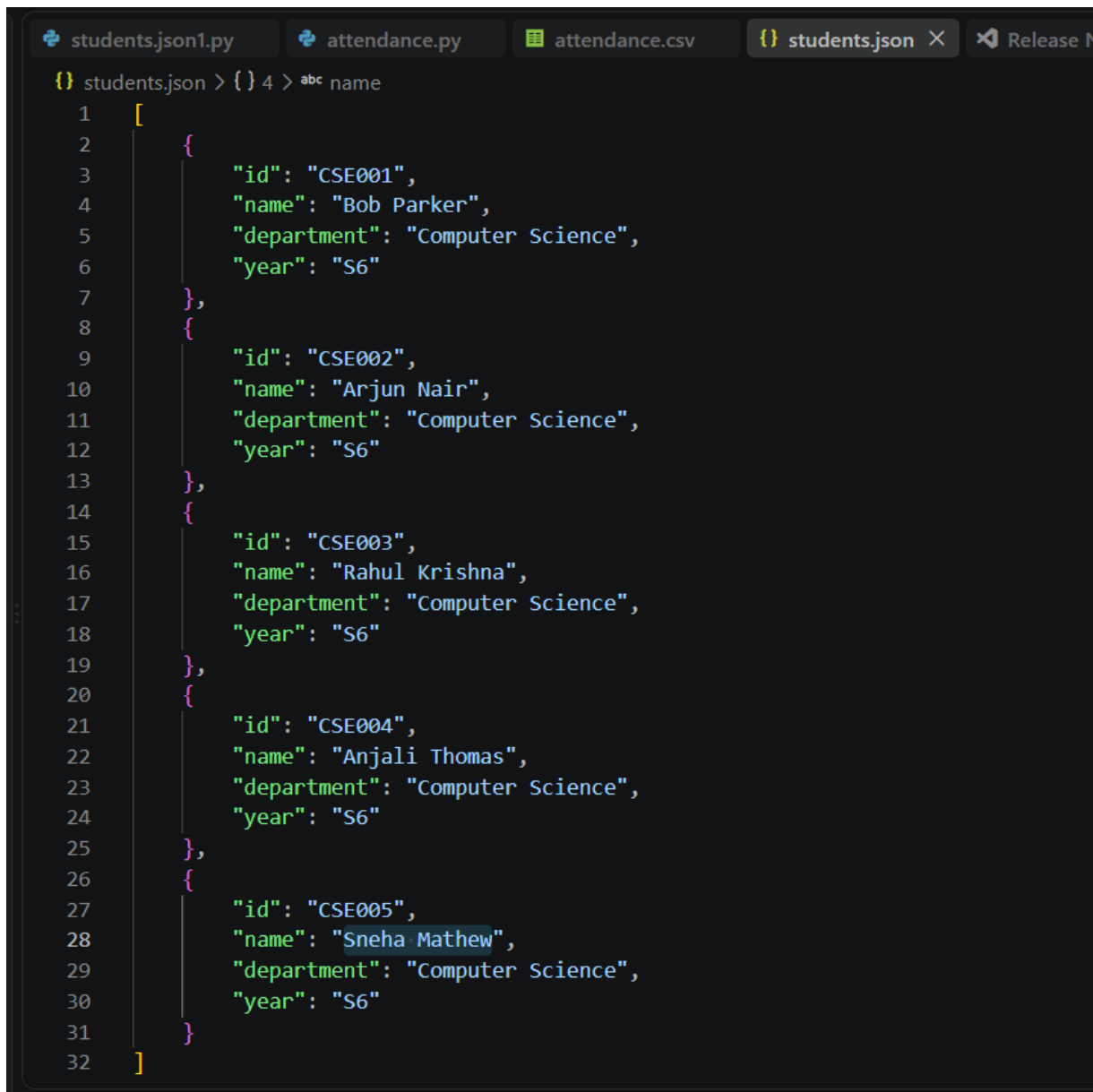
The simulation begins by loading the AI Attendance Agent modules and initializing the student database. The user enters the detected student's name, simulating facial recognition.

If the student exists in the database, the system displays AI analysis, including:

- Student Information
- Confidence Score
- Liveness Status
- Mask Detection Status

Attendance is then stored in the CSV file. Unknown students are rejected, while duplicate attendance is prevented. Finally, the application displays an attendance dashboard summarizing the session

9 Screenshots of Full Code



```
students.json1.py  attendance.py  attendance.csv  {} students.json  X  Release N

{} students.json > {} 4 > abc name
1  [
2      {
3          "id": "CSE001",
4          "name": "Bob Parker",
5          "department": "Computer Science",
6          "year": "S6"
7      },
8      {
9          "id": "CSE002",
10         "name": "Arjun Nair",
11         "department": "Computer Science",
12         "year": "S6"
13     },
14     {
15         "id": "CSE003",
16         "name": "Rahul Krishna",
17         "department": "Computer Science",
18         "year": "S6"
19     },
20     {
21         "id": "CSE004",
22         "name": "Anjali Thomas",
23         "department": "Computer Science",
24         "year": "S6"
25     },
26     {
27         "id": "CSE005",
28         "name": "Sneha Mathew",
29         "department": "Computer Science",
30         "year": "S6"
31     }
32 ]
```

Figure 1: Screenshot of Student Database

```

attendance.py
56
57 while True:
58
59     print("\nScanning Classroom...")
60     time.sleep(1)
61
62     print("Face Detected\n")
63
64     name = input("Enter Detected Student Name (or exit): ")
65
66     if name.lower() == "exit":
67         break
68
69     student = None
70
71     for s in students:
72         if s["name"].lower() == name.lower():
73             student = s
74             break
75
76     if student:
77
78         if student["id"] in marked:
79
80             print("\nDuplicate Entry Detected")
81             print("Attendance already recorded.\n")
82             continue
83
84         confidence = round(random.uniform(95, 99.9), 2)
85
86         liveness = random.choice(["PASS", "PASS", "PASS", "PASS"])
87
88         mask = random.choice(["No Mask", "Mask Detected"])
89
90         print("\nSearching Database...")
91         time.sleep(1)
92

```

Figure 2 :Screenshot of Face Recognition and Student Matching

10 Screenshots of Output

```
PS C:\Users\Nithin Jose\Desktop\AI_Assignment> py attendance.py

  AI SMART ATTENDANCE AGENT v2.0
  Facial Recognition Simulation Engine

Loading Camera Module...
Loading Face Recognition Engine...
Loading Student Database...
Loading Attendance Database...
Loading AI Decision Module...

System Status : ONLINE
-----

Scanning Classroom...
Face Detected

Enter Detected Student Name (or exit):
```

Figure 3: Screenshot of System Initialization

```
Enter Detected Student Name (or exit): Bob parker

Searching Database...

Match Found

Student Information
-----
ID          : CSE001
Name        : Bob Parker
Department  : Computer Science
Year        : S6

AI Analysis
-----
Confidence   : 97.03 %
Liveness     : PASS
Mask Detection : No Mask

Decision
-----
Attendance Accepted

Scanning Classroom...
Face Detected

Enter Detected Student Name (or exit): █
```

Figure 4 :Screenshot of Successful Attendance

```
Enter Detected Student Name (or exit): Bob Parker  
  
Duplicate Entry Detected  
Attendance already recorded.  
  
Scanning Classroom...  
Face Detected
```

Figure 5 :Screenshot of Duplicate Attendance Detection

```
Enter Detected Student Name (or exit): John James  
  
WARNING  
-----  
Unknown Face  
Attendance Denied  
  
Scanning Classroom...  
Face Detected
```

Figure 6 :Screenshot of Unknown Face Detection

```
PS C:\Users\Nithin Jose\Desktop\AI_Assignment> python attendance.py
=====
TODAY'S ATTENDANCE DASHBOARD
=====
✓ Arjun Nair
✓ Rahul Krishna
✓ Bob Parker

Total Students : 5
Present        : 3
Absent         : 2
Attendance %   : 60.0 %

Attendance Saved Successfully
System Closed
PS C:\Users\Nithin Jose\Desktop\AI_Assignment> 
```

Figure 7 : Screenshot of Attendance Report

```
students.json1.py  attendance.py  README.md  attendance.csv M X  students.json
attendance.csv
1 CSE001,Bob Parker,Computer Science,06-08-2026,21:57:28,97.83,PASS,Mask Detected,Present
2 CSE002,Arjun Nair,Computer Science,06-08-2026,21:59:15,98.11,PASS,Mask Detected,Present
3 CSE003,Rahul Krishna,Computer Science,06-08-2026,21:59:21,97.56,PASS,Mask Detected,Present
4 CSE004,Anjali Thomas,Computer Science,06-08-2026,21:59:28,96.73,PASS,Mask Detected,Present
5 CSE005,Sneha Mathew,Computer Science,06-08-2026,21:59:41,96.48,PASS,No Mask,Present
```

Figure 8 : Screenshot of Generated Attendance Record (CSV File)